

Abstract No.

Liquid Crystal-like Self-Organization of Glioblastoma is Associated with Consistent Cell Migration

Urszula Hohmann¹, Chalid Ghadban¹, Julian Prell², Christian Strauss², Faramarz Dehghani¹, and Tim Hohmann^{1,}*

¹Department of Anatomy and Cell Biology, Martin Luther University Halle-Wittenberg

²Department of Neurosurgery, Martin Luther University Halle-Wittenberg

*Presenting author

Glioblastoma is the most lethal and frequent type of primary brain tumors, characterized by a high proliferative and infiltrative capacity. Here, we used live cell imaging to analyze the effect of cell density variations on the migratory capacity of established and primary glioblastoma cell lines. We found that proliferation events promoted local velocity of glioblastoma cells, up to three cell-length away from the proliferation event. Furthermore, two phenotypes were found when subjecting glioblastoma cells to a cell density gradient. While one phenotype was characterized by the active migration of cells, independent of proliferation, the other was mostly driven by cell proliferation. Lastly, the analysis of effects of an overall increasing cell density demonstrated that cells showing signs of self-organization, forming liquid crystal-like structures are able to maintain a high migratory potential even for high cell densities, normally considered to result in a migratory arrest. Notably, the emergence of small-scale liquid crystal-like order was associated with a better maintenance of cellular migration, even in cell populations that were otherwise largely in a state of migratory arrest. Thus, structure formation might help glioblastoma cells to move efficiently in states of high confinement, maintaining infiltrative properties.



Supported by SciSerTec

Web: <http://www.scisertec.de>